

Normalization

IEM 4723 Information Systems Design

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What Normalization Is

Introduction

- Normalization is a process for arriving at a good relational design.
- An ER diagram is created first, and that diagram is used to design the relations. Those relations are then checked against a set of theoretical rules for good design, and changed if they do not meet them.
- This follows the same ER modeling process covered earlier in the course. First the context data model, then the key-based data model, then the fully-attributed data model, and finally the entities become relations.
- The theoretical rules that the design of a relation must meet are called normal forms.

The Normal Forms

- There are six normal forms, 1NF, 2NF, 3NF, BCNF, 4NF, and 5NF.
- The higher the normal form, the better the design of the relation.
- Relations in 3NF already avoid most of the problems common to bad relational designs. The higher normal forms handle special situations that arise only occasionally.
- For that reason this course concentrates on the first three.

The First Three Normal Forms

First Normal Form

- First normal form is about single-valued attributes and the absence of repeating groups.
- A 1NF entity is an entity whose attributes have no more than one value for a single instance of that entity. There should also be no repeating groups.
- An attribute that can hold multiple values, or that repeats, is describing a separate entity, and possibly an entity together with a relationship.

Source: Whitten and Bentley (2007)

Second Normal Form

- Second normal form is about partial dependency, specifically the absence of it.
- A 2NF entity is an entity whose non-key attributes depend on the full primary key.
- An entity with a single-attribute primary key cannot have a partial dependency, so it is already in 2NF.
- Any non-key attribute that depends on only part of the primary key should be moved to an entity where that partial key is the full key. This may require creating a new entity and a new relationship in the data model.

Source: Whitten and Bentley (2007)

Third Normal Form

- Third normal form is about transitive dependency, again the absence of it.
- A 3NF entity is an entity whose non-key attributes do not depend on any other non-key attribute.
- Any non-key attribute that depends on another non-key attribute must be moved or deleted. Here too, new entities and relationships may have to be added to the data model.

Source: Whitten and Bentley (2007)

Repeating Groups in Practice

A Table With Repeating Groups

emp#	first	last	children's names	children's birthdates
1001	Jane	Doe	Mary, Sam	1/9/02, 5/15/04
1002	John	Doe	Lisa, David	1/9/00, 5/15/01
1003	Jane	Smith	John, Pat, Lee, Mary	10/5/04, 10/12/00, 6/6/2006, 8/21/04
1004	John	Smith	Michael	7/4/06
1005	Jane	Jones	Edward, Martha	10/21/05, 10/15/99

Fig. 7.2 A table with repeating groups

- A relation in 1NF has no repeating groups, which are the analogue of multivalued attributes in an ER diagram.
- This table has two major problems. It is difficult to associate each child with the right birthdate, and it is difficult to search, for example to find every employee with a child named John.

Handling Repeating Groups, the Wrong Way

emp#	first	last	child name 1	child bdate 1	child name 2	child bdate 2	child name 3	child bdate 3
1001	Jane	Doe	Mary	1/1/02	Sam	5/15/04		
1002	John	Doe	Lisa	1/1/00	David	5/15/01		
1003	Jane	Smith	John	10/5/04	Pat	10/12/00	Lee	6/6/06
1004	John	Smith	Michael	7/4/06				
1005	Joe	Jones	Edward	10/21/05	Martha	10/15/99		

Fig. 7.3 A relation handling repeating groups in the wrong way

- One tempting fix is to create multiple columns to hold the multiple values.
- This raises the question of how many columns to create, it wastes space whenever an employee has fewer children than there are columns, and searching is still difficult, since finding a child named John now means searching all three name columns.

Handling Repeating Groups, the Correct Way

employee

emp#	first	last
1001	Jane	Doe
1002	John	Doe
1003	Jane	Smith
1004	John	Smith
1005	Joe	Jones

children

emp#	c_first	c_birthdate
1001	Mary	1/1/02
1001	Sam	5/15/04
1002	Lisa	1/1/00
1002	David	5/15/01
1003	John	10/5/04
1003	Pat	10/12/00
1003	Lee	6/6/06
1003	Mary	8/21/04
1004	Michael	7/4/06
1005	Edward	10/21/05
1005	Martha	10/15/99

- The correct approach is to create another table to hold the multiple instances of the repeating group.
- The employee details stay in one relation, and each child becomes its own row in a second relation, linked back by the employee number.
- Searching for a child named John is now a single condition on a single column, and every child sits alongside its own birthdate.

Fig. 7.4 The correct way to handle a repeating group

A Worked Example

The Res_Appt Relation

Consider the following Res_Appt table, designed to hold details of graduate student research appointments in an IE department.

Res_Appt (StuID, StuName, ResAccNum, StartDate,
EndDate, SalCode, EMailID, FTEPercent, MoRate)

- Before it can be normalized, we need to know which attributes determine which others. Those relationships are the functional dependencies of the relation.

The Functional Dependencies

FD1: StuID $\rightarrow$ StuName, EMailID

FD2: StuID + ResAccNum $\rightarrow$ StartDate, EndDate, SalCode

FD3: SalCode $\rightarrow$ FTEPercent, MoRate

- Read each of these as the attributes on the left determining the attributes on the right.
- Notice that FD2 has two attributes on its left side, while FD1 and FD3 each have only one. That difference is what the normalization work turns on.

Questions

- 1 Determine the primary key of the Res_Appt relation.
- 2 Decompose Res_Appt into 2NF relations, and then into 3NF relations. Clearly identify the primary key of each newly formed relation.